



GATE SMASHERS

GATE2024

Engineering Mathematics

Topic: Probability and Statistics

Subtopic:**Binomial Distribution**.....

LECTURE - 18

EXPLANATION

Binomial Distribution - Discovered by Swiss Mathematician James Bernoulli in 1700, hence it is also known as Bernoulli Distribution.



Binomial Distribution $\begin{cases} p \text{ (success)} \\ q \text{ (failure)} \end{cases}$

Ideas -

- There should be finite no. of trials.
- Trial should be independent.
- Trial should have exactly two outcomes
- Probability of success remains same in each trial.

Formula -

$$P(X=r) = {}^nC_r p^r q^{n-r}$$

n → no. of trial
 r → Random Variable
 p → probability of success
 q → probability of failure

n & p → Parameters of binomial distribution.

Mean (μ) - np

Variance (σ^2) - npq → $np > npq$

Standard Deviation (σ) - $\sqrt{npq}$

- $p = q = \frac{1}{2} \rightarrow$ Symmetrical Distribution
- $p \neq q \rightarrow$ Skewed Distribution

NOTES PREPARED BY: ANKIT ARORA



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EXPLANATION

Example 1) - An unbiased coin is tossed 10 times.
Find using binomial distribution



- i) Exactly 6 heads.
- ii) Atleast 6 heads.
- iii) Atmost 6 heads.
- iv) Atleast 3 heads.

Example 2- A pair of dice is thrown 7 times. If getting
a total 7 is considered as success. What is
the probability of

- | | |
|------------------|---------------------------|
| (i) No success | (iii) atleast 6 successes |
| (ii) 6 successes | (iv) atmost 6 successes. |

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